

Titanic Dataset - Exploratory Data Analysis (EDA)

1. Introduction

Exploratory Data Analysis (EDA) was performed on the Titanic dataset to understand data patterns, relationships, trends, and anomalies. The analysis was conducted using Python libraries such as Pandas, Matplotlib, and Seaborn.

2. Dataset Overview

The Titanic dataset contains information about passengers, including age, gender, passenger class, fare, and survival status.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

3. Data Understanding

Dataset Information

The dataset was examined using:

- `df.info()`
- `df.describe()`
- `df.value_counts()`

Missing Values

Missing values were identified and handled using median imputation for numerical columns such as Age and Fare.

4. Univariate Analysis

Survival Distribution

A count plot was used to analyze survival status.

Observation:

Most passengers did not survive. The survival rate was lower than the non-survival rate.

Age Distribution

A histogram was used to visualize age distribution.

Observation:

Most passengers were between 20 and 40 years old.

Fare Distribution

A histogram was used to analyze fare distribution.

Observation:

Fare distribution was positively skewed, with a small number of passengers paying significantly higher fares.

5. Bivariate Analysis

Gender vs Survival

A count plot was used to compare survival based on gender.

Observation:

Female passengers had a higher survival rate compared to male passengers.

Passenger Class vs Survival

A count plot was used to compare survival across passenger classes.

Observation:

Passengers traveling in First Class had a higher survival rate than those in Second and Third Class.

Age vs Fare

A scatter plot was used to examine the relationship between Age and Fare.

Observation:

No strong relationship was observed between Age and Fare.

6. Multivariate Analysis

Correlation Heatmap

A heatmap was used to identify correlations among numerical variables.

Observation:

Most variables showed weak to moderate correlation. No severe multicollinearity was observed.

Pairplot

A pairplot was used to visualize relationships among Age, Fare, Passenger Class, and Survival.

Observation:

The pairplot provided an overall understanding of variable distributions and

relationships.

7. Key Findings

1. Most passengers were between 20 and 40 years old.
2. Female passengers had significantly higher survival rates.
3. First-class passengers survived more frequently.
4. Fare distribution was positively skewed.
5. Several fare outliers were identified.
6. No strong correlation existed among most numerical variables.
7. Passenger class and gender were important factors influencing survival.

8. Conclusion

The Exploratory Data Analysis of the Titanic dataset revealed meaningful insights regarding passenger survival patterns. Gender and passenger class were the most influential factors affecting survival. EDA helped identify trends, detect outliers, and understand relationships among variables, providing a strong foundation for future predictive modeling.

References

- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Titanic Dataset